```
clc;
clear;
close all;

%% VSB-SC MODULATOR AND DEMODULATOR

%% 1. Parameters
Fs = 500000;
T = 0.02;
t = 0:1/Fs:T-1/Fs;
N = length(t);

fc = 100000;       % Carrier frequency
Ac = 1;            % Carrier amplitude

fm1 = 2000;
fm2 = 5000;
fm3 = 8000;

%% 2. Message / TV Video Signal
m = 0.5*cos(2*pi*fm1*t) + ...
    0.3*cos(2*pi*fm2*t) + ...
    0.2*cos(2*pi*fm3*t);

m = m / max(abs(m));

%% 3. Carrier
carrier = Ac*cos(2*pi*fc*t);

%% 4. DSB-SC Modulation
dsb_sc = m .* carrier;

%% 5. Frequency Axis
f = (-N/2:N/2-1)*(Fs/N);

%% 6. DSB-SC Spectrum
DSB = fftshift(fft(dsb_sc));

%% 7. VSB Filter
Bm = fm3;
Bv = 2000;

H = zeros(size(f));

% Positive frequencies
H(f >= fc & f <= fc + Bm) = 1;
H(f >= fc-Bv & f < fc) = 0.5;

% Negative frequencies
H(f <= -fc & f >= -fc-Bm) = 1;
```

```
H(f < -fc & f >= -fc+Bv) = 0.5;

%% 8. VSB Signal
VSB_spectrum = DSB .* H;

vsb = real(ifft(ifftshift(VSB_spectrum)));

%% 9. Add Noise
SNR_dB = 30;

signal_power = mean(vsb.^2);
noise_power = signal_power / (10^(SNR_dB/10));

noise = sqrt(noise_power) * randn(size(vsb));

received_vsb = vsb + noise;

%% 10. Coherent Detection
local_carrier = 2*cos(2*pi*fc*t);

detector_output = received_vsb .* local_carrier;

%% 11. Low Pass Filter
cutoff = Bm;

[b,a] = butter(6, cutoff/(Fs/2));

recovered_signal = filtfilt(b,a,detector_output);

%% 12. Normalize
recovered_signal = recovered_signal / max(abs(recovered_signal));

%% 13. Performance Analysis

MSE = mean((m - recovered_signal).^2);

R = corrcoef(m,recovered_signal);
Correlation = R(1,2);

signal_power = mean(m.^2);
error_power = mean((m-recovered_signal).^2);

SNR = 10*log10(signal_power/error_power);

%% 14. Display Results

fprintf('\n====================================\n');
```

```
==================================
```

```
fprintf('       VSB SYSTEM RESULTS\n');
```

```
       VSB SYSTEM RESULTS
```

```
fprintf('====================================\n');
```

```
 ==================================
```

```
fprintf('Carrier Frequency       = %.2f kHz\n',fc/1000);
```

```
 Carrier Frequency       = 100.00 kHz
```

```
fprintf('Message Bandwidth       = %.2f kHz\n',Bm/1000);
```

```
 Message Bandwidth       = 8.00 kHz
```

```
fprintf('Vestigial Bandwidth     = %.2f kHz\n',Bv/1000);
```

```
 Vestigial Bandwidth     = 2.00 kHz
```

```
fprintf('VSB Bandwidth           = %.2f kHz\n',(Bm+Bv)/1000);
```

```
 VSB Bandwidth           = 10.00 kHz
```

```
fprintf('\nPerformance:\n');
```

```
 Performance:
```

```
fprintf('MSE                     = %.6f\n',MSE);
```

```
 MSE                     = 0.052994
```

```
fprintf('Correlation Coefficient = %.4f\n',Correlation);
```

```
 Correlation Coefficient = 0.9760
```

```
fprintf('SNR                     = %.2f dB\n',SNR);
```

```
 SNR                     = 5.55 dB
```

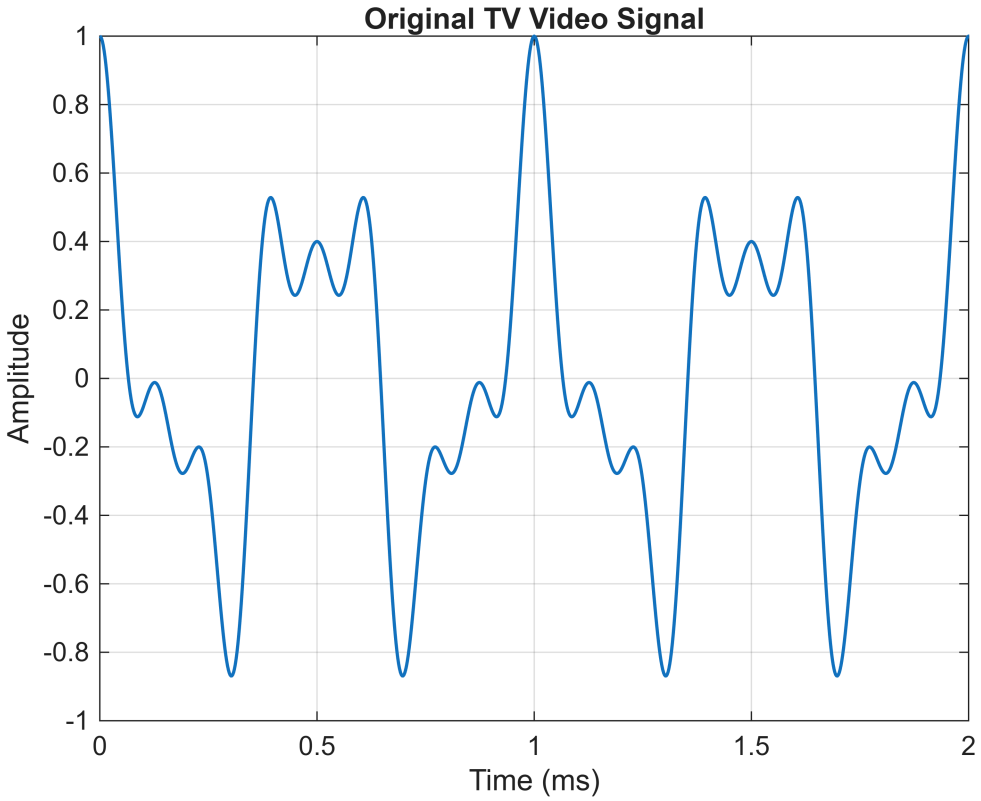
```
fprintf('====================================\n');
```

```
 ==================================
```

```
%% 15. Plot Original Signal

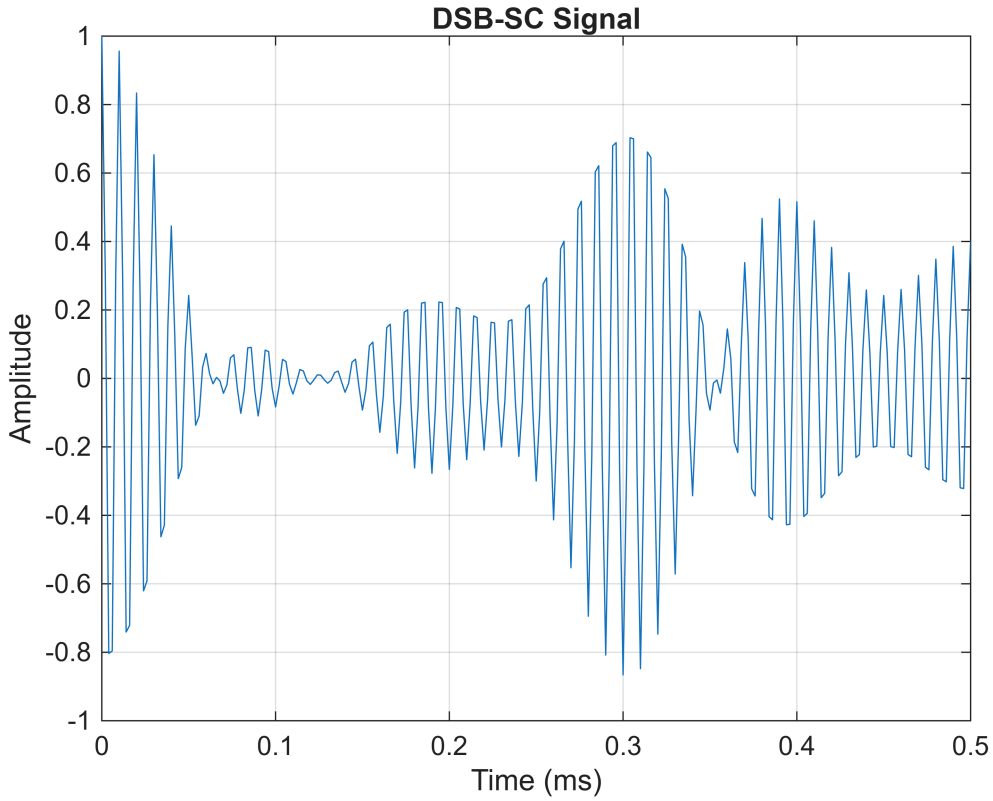
figure;
plot(t*1000,m,'LineWidth',1.2);
xlabel('Time (ms)');
ylabel('Amplitude');
title('Original TV Video Signal');
grid on;
```

```
xlim([0 2]);
```



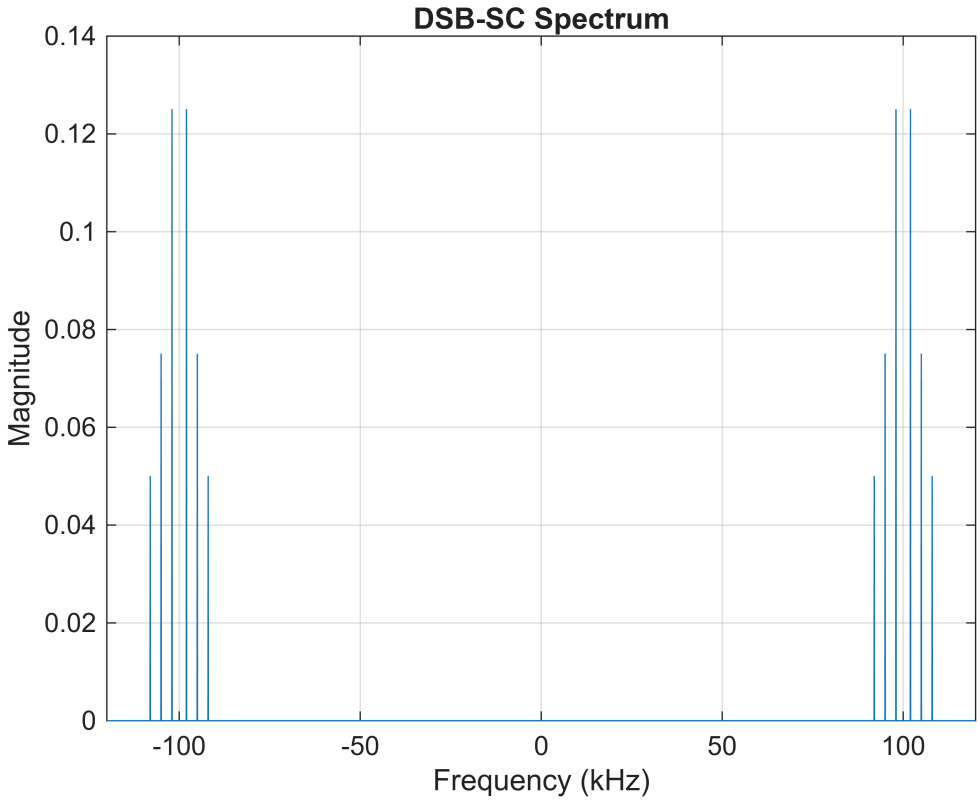

```
%% 16. Plot DSB-SC Signal

figure;
plot(t*1000,dsb_sc);
xlabel('Time (ms)');
ylabel('Amplitude');
title('DSB-SC Signal');
grid on;
xlim([0 0.5]);
```



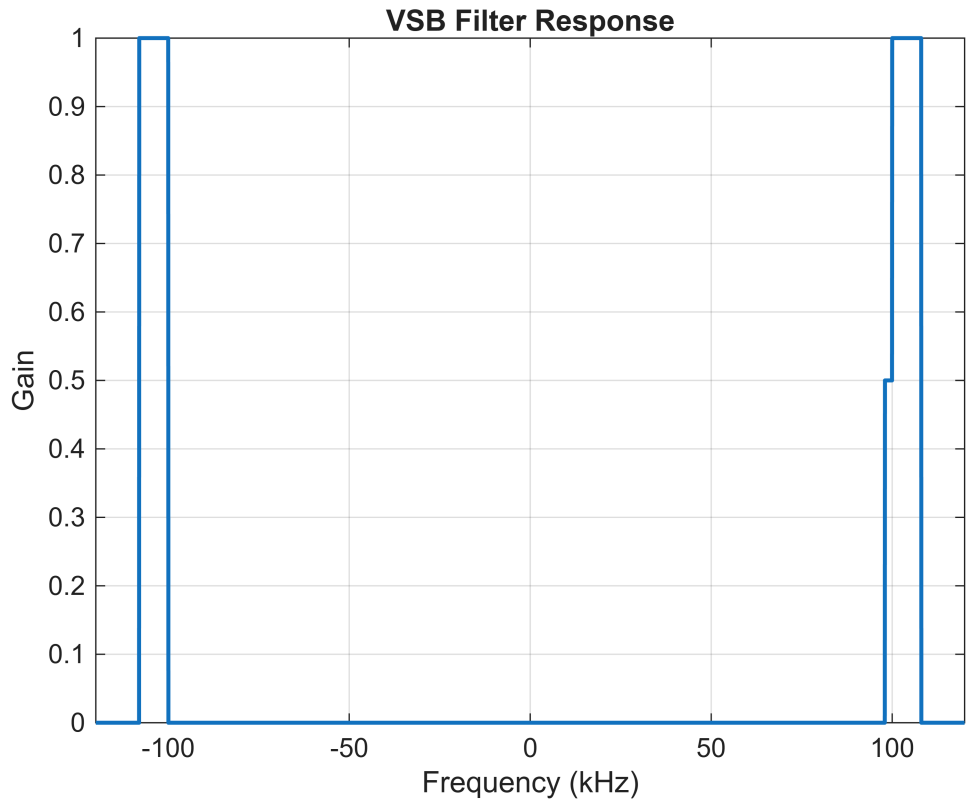

```
%% 17. DSB-SC Spectrum

figure;
plot(f/1000,abs(DSB)/N);
xlabel('Frequency (kHz)');
ylabel('Magnitude');
title('DSB-SC Spectrum');
grid on;
xlim([-120 120]);
```



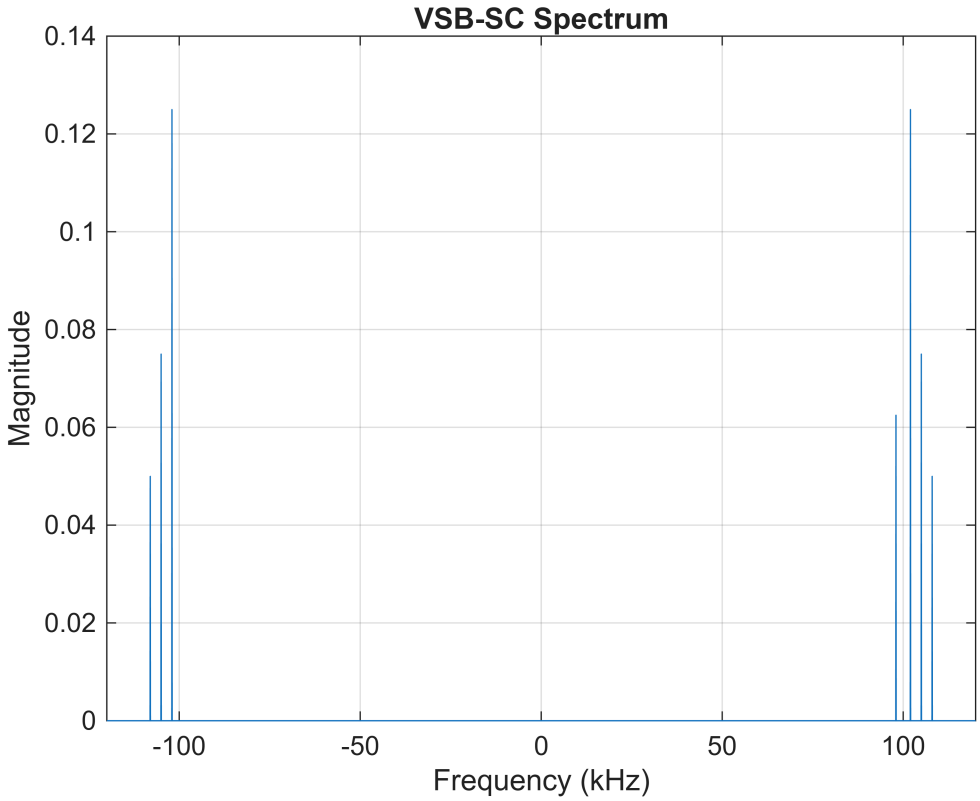

```
%% 18. VSB Filter Response

figure;
plot(f/1000,H,'LineWidth',1.5);
xlabel('Frequency (kHz)');
ylabel('Gain');
title('VSB Filter Response');
grid on;
xlim([-120 120]);
```



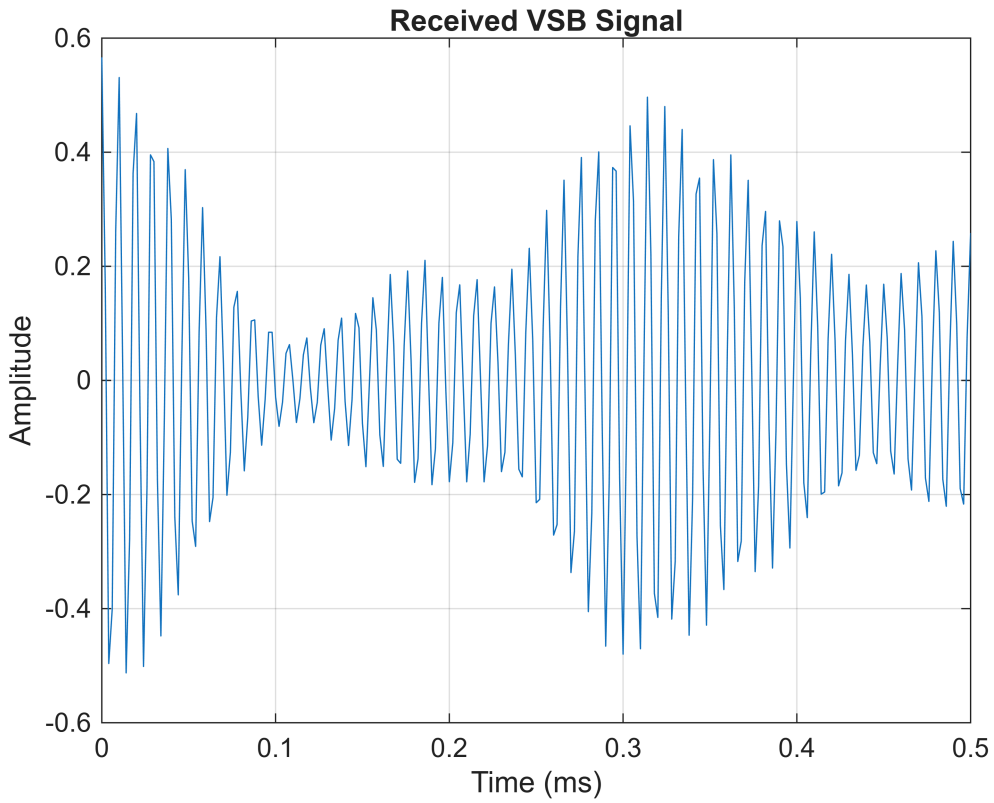

```
%% 19. VSB Spectrum

figure;
plot(f/1000,abs(VSB_spectrum)/N);
xlabel('Frequency (kHz)');
ylabel('Magnitude');
title('VSB-SC Spectrum');
grid on;
xlim([-120 120]);
```



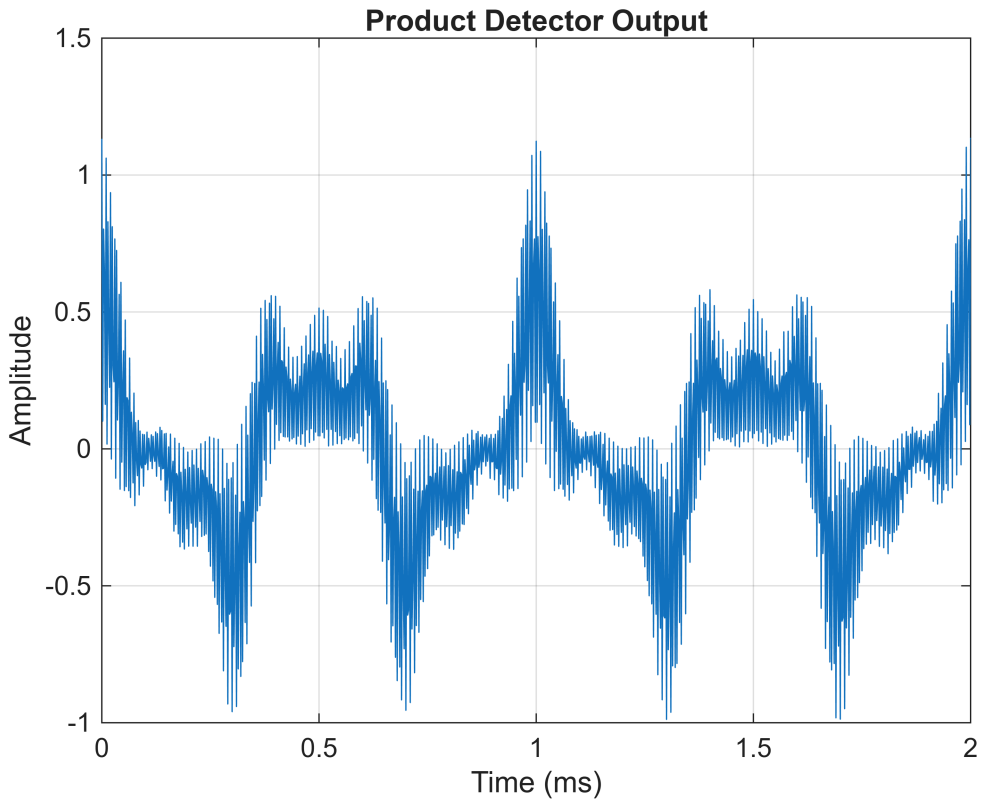

```
%% 20. Received VSB Signal

figure;
plot(t*1000,received_vsb);
xlabel('Time (ms)');
ylabel('Amplitude');
title('Received VSB Signal');
grid on;
xlim([0 0.5]);
```



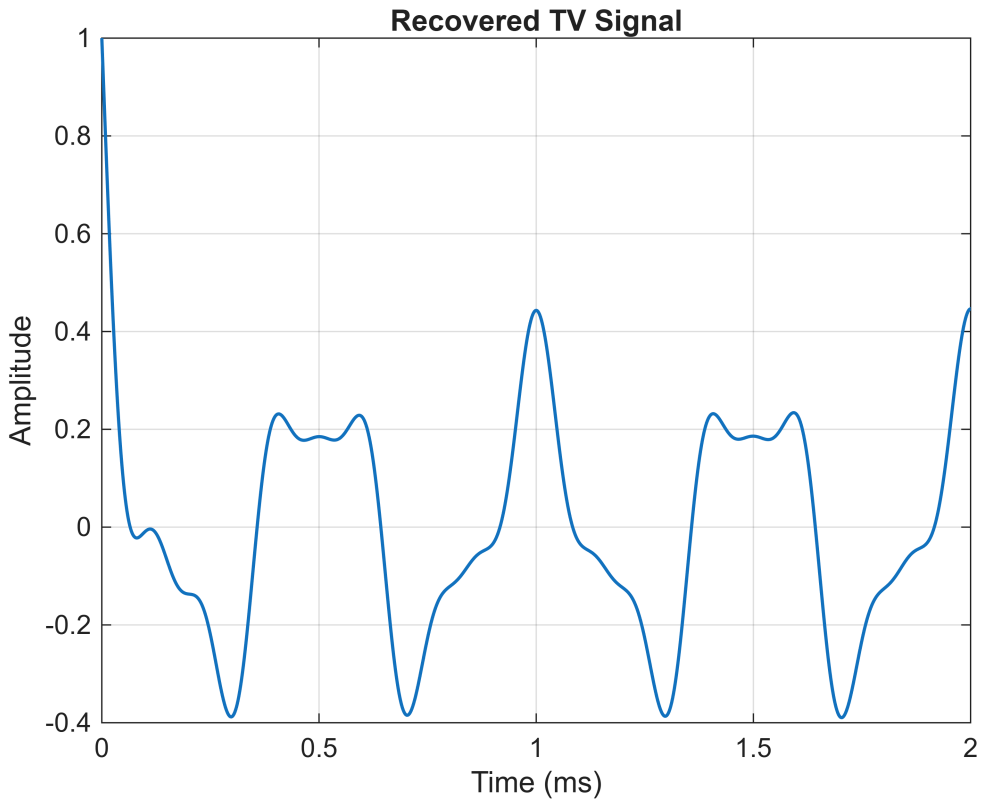

```
%% 21. Product Detector Output

figure;
plot(t*1000,detector_output);
xlabel('Time (ms)');
ylabel('Amplitude');
title('Product Detector Output');
grid on;
xlim([0 2]);
```




```
%% 22. Recovered Signal

figure;
plot(t*1000,recovered_signal,'LineWidth',1.2);
xlabel('Time (ms)');
ylabel('Amplitude');
title('Recovered TV Signal');
grid on;
xlim([0 2]);
```




```
%% 23. Original vs Recovered

figure;

plot(t*1000,m,'LineWidth',1.2);
hold on;

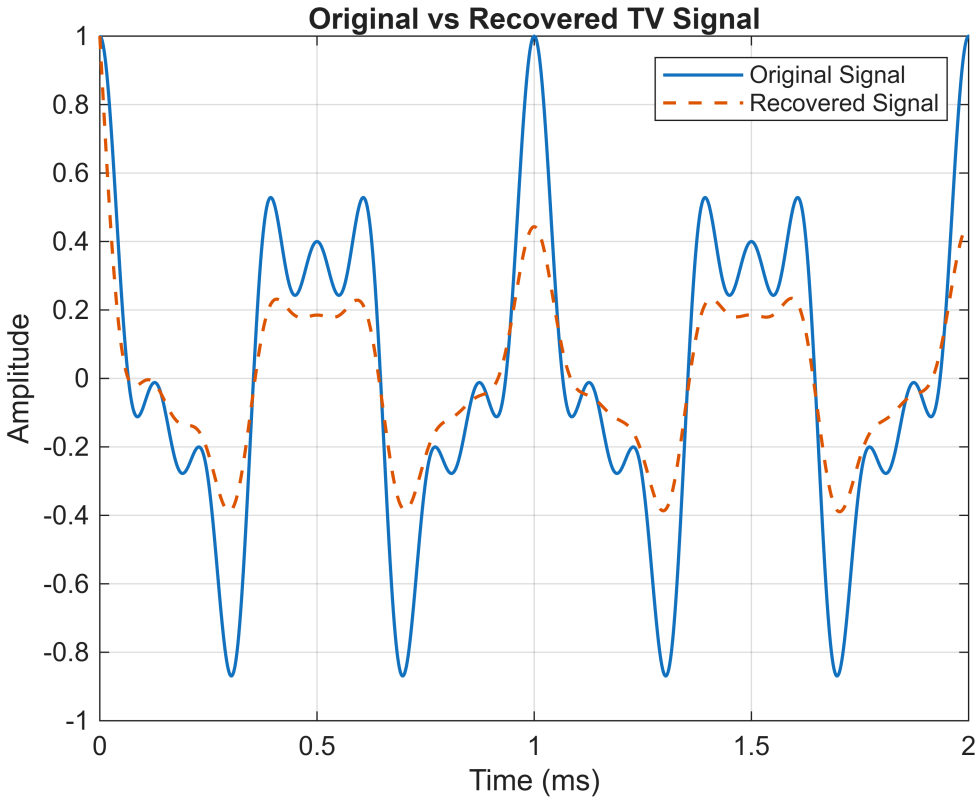
plot(t*1000,recovered_signal,'--','LineWidth',1.2);

xlabel('Time (ms)');
ylabel('Amplitude');

title('Original vs Recovered TV Signal');

legend('Original Signal','Recovered Signal');

grid on;
xlim([0 2]);
```


Original vs Recovered TV Signal
Original Signal
Recovered Signal
Amplitude
Time (ms)
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
0
0.5
1
1.5
2